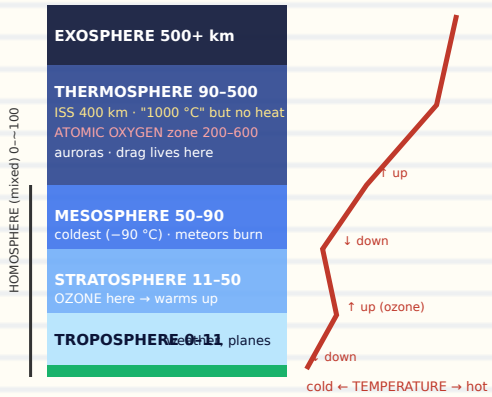
SEC 2 · LEC 5 · EARTH ATMOSPHERE

Five layers, one zigzag, one hungry atom

Three slides: the layers, what the air is made of at each height, and atomic oxygen. One drawing covers all three: the layer stack with the temperature zigzag and who lives where.

🖊 LEFT · DRAW

S2 / L5 — Earth atmosphere **11/8 · p__**



⭐ **3 TAKEAWAYS**

1. Temp zigzags: down, up (ozone eats UV → heat), down, way up
2. Thermosphere is "hot" but too thin to burn you: temp ≠ heat
3. Atomic O = UV-split oxygen; dominant 200–600 km; s/c hits it at 7.5 km/s

🧠 **SUMMARY · video closed · 3 lines**

1. _____
2. _____
3. _____

❓ RIGHT · CORNELL

CUES	NOTES
5 layers, bottom to top, with km?	tropo 0–11 · strato 11–50 · meso 50–90 · thermo 90–500 · exo 500+
Why does temp go UP in the stratosphere?	temp: ↓ ↑ ↓ ↑↑ · strato ↑ b/c ozone absorbs UV → heat (L4 callback)
Thermosphere is 1000 °C. Would the ISS melt? Why not?	thermo "500–1500 °C" but air ~nothing ∴ no heat to transfer (no convection, L4)
What's the air made of low down vs at 300 km?	homosphere 0–~100 km = gases mixed (N_2 78%, O_2 21%, Ar) above ~100: gases sort by weight · chart is log scale (each line = 100,000×)
Atomic O: what, where from, where does it live?	atomic O (single atom, not O_2) = UV splits O_2 → dominant gas ~200–600 km = LEO
Why does 7.5 km/s matter if the atoms barely move?	very reactive → oxidation eats polymers (Kapton), metals slower atoms ~still; s/c slams into them at 7.5 km/s → ~5 eV per hit > bond energy
Metals vs polymers under atomic O?	fix: coatings (SiO_2, Al), AO-resistant materials, shield ram-facing side
? what is "anomalous O"	corrosion IS the right word here (oxidation), unlike UV in L4

How to use this

- **Draw the stack first.** Five bands, heights on the left, the red temperature zigzag on the right. Then put the ISS and the atomic-oxygen zone in the thermosphere band.
- Notes column is at the limit again. Cut what you already knew.
- Two callbacks to last lecture: ozone + UV, and no convection. Green-highlight those on review if you can explain the link.
- Page 2: the captions mangled 8 words. The list is there so you don't copy a wrong one.

⚡ Slide 1 (the title) wasn't sent. Fine. The three you sent are the whole content.

SEC 2 · LEC 5 · TEACH + FACT CHECK

The air gets thinner, weirder, and hungrier as you go up

Plain-English version, then the caption errors and the two places the instructor was loose. Read on screen; don't copy.

🪜 The stack, floor by floor

- **Troposphere, 0–11 km.** Everything you've ever experienced: weather, clouds, airliners at 10 km. Gets colder as you climb, about −50 °C at the top.
- **Stratosphere, 11–50 km.** Temperature goes *up* with height, which feels backwards. Reason: the ozone layer lives here and it soaks up UV. Absorbed light becomes heat. Last lecture's "sunscreen" is also a heater.
- **Mesosphere, 50–90 km.** Coldest place on the whole stack, about −90 °C. Meteors burn up here.
- **Thermosphere, 90–500 km.** Temperature shoots up to 500–1500 °C. But the air is so thin there's almost nothing to carry that heat into you. A thermometer reads hot; your hand would feel nothing and the ISS, which orbits right in the middle of this layer at 400 km, does not melt. Temperature is how fast the few molecules move; heat is how much energy they can hand over. Here, almost none.
- **Exosphere, 500 km and up.** So thin it blends into space. Atoms here can escape Earth entirely.
- **Homosphere** is the instructor's "first region": the bottom ~100 km where the gases are stirred together so the recipe is the same everywhere (78% nitrogen, 21% oxygen, 1% argon). Above that, gases separate out by weight, heavy ones low, hydrogen and helium highest.

🦷 Atomic oxygen, the hungry atom

- Normal oxygen is two atoms holding hands: O_2. Up high, UV light (no ozone to stop it) snaps the pair apart into single O atoms. A single O atom is desperate to bond with anything. That's "highly reactive."
- The chart shows it: the brown line (O) rises above the N_2 line around 200 km and stays on top. From roughly 200 to 600 km, atomic oxygen is the main gas. That is exactly where LEO satellites fly.
- The atoms are nearly stationary. The spacecraft is the thing moving, at 7.5 km/s. So every atom hits the front face like a bullet, with about 5 eV of impact energy. More than enough to break a bond.
- Polymers like Kapton erode visibly in months (the slide photo). Metals form a thin oxide skin and then mostly stop. Fixes: thin glass or aluminum coatings, oxygen-resistant materials, and keeping delicate stuff off the side that faces forward.

🔎 Caption errors (don't copy these)

Captions said	Instructor meant
"the comet"	the **homosphere**
"density and competence"	density and **components**
"the great of a time"	**degrade** over time
"corrosion and explosion"	corrosion and **erosion**
"destroying the bones"	destroying the **bonds**
"polymers are followable"	polymers are **vulnerable**
"increasing the acceleration rate"	increasing the **oxidation** rate
"320–340 comets"	320–340 **km** (see below)

▲ Where the instructor was loose

- **"Atomic oxygen appears around 320–340 km."** Not quite. His own chart shows atomic O present from the left edge (80 km) and dominant from ~200 km. What appears at 320–340 km on that chart is the black "anomalous O" line, a minor extra form. Write: atomic O dominant ~200–600 km.
- **"Mainly only atomic nitrogen increases."** Atomic N does rise, but it's a minor player. The big story on the chart is atomic O taking over from N_2 and O_2. Don't let the nitrogen line steal the lesson.
- **"Corrosion" is correct this time.** Last lecture he used it loosely for UV. Here atomic oxygen really does oxidize materials, which is what corrosion means. Same word, now accurate.
- **Not mentioned at all: drag.** That thin thermosphere air also slows satellites down. The ISS loses ~2 km of altitude a month and has to fire thrusters to climb back. It'll show up later; note it now.

📒 Question bank cards

- Order the 5 layers and give the height where each starts.
- Why is the stratosphere warmer at the top than the bottom?
- The thermosphere is 1000 °C. Why doesn't the ISS cook?
- How is atomic oxygen made, and where is it the main gas?

SEC 2 · LEC 5 · VOCABULARY

Every word used, in plain English

Includes the words on the chart slide, which the instructor didn't explain at all. Copy only the ones that were new to you into the word bank, 3–5 word meaning, with the page number.

Atmosphere — the blanket of gas held around a planet by gravity. Thick at the bottom, fading to nothing a few hundred km up. *Earth's air, all of it.*

Troposphere — bottom floor, 0–11 km. Weather, clouds, airliners. *"Tropo" = turning: the air churns here.*

Mesosphere — 50–90 km. The coldest layer; meteors burn up here. *"Meso" = middle.*

Exosphere — 500 km and beyond. So thin that atoms can drift off into space. *"Exo" = outside.*

Temperature vs heat — temperature = how fast molecules move. Heat = how much energy they can actually give you, which also depends on how many there are. *One sparkler is 1000 °C and won't warm a room.*

Log scale — the chart's vertical axis where each tick (10^{10}, 10^{15}, 10^{20}) is 100,000× the one below. Used when numbers span huge ranges. *Zooming out so a mountain and a pebble fit on one page.*

N_2, O_2, Ar — nitrogen gas (two N atoms), oxygen gas (two O atoms), argon. The three main ingredients of normal air: 78%, 21%, 1%.

Atomic oxygen (AO) — a single O atom on its own, made when UV splits O_2. Craves a bond with anything, so it eats materials. Main gas from ~200–600 km. *A loose dog with no leash.*

NRLMSISE-00 — just the name of the computer model that produced the chart (a US Navy atmosphere model). Not a thing to learn; it's a citation.

VLEO / LEO — Very Low Earth Orbit (below ~250 km, heavy drag, short life) vs Low Earth Orbit (up to 2,000 km). The green line on the chart marks the split.

Oxidation / corrosion — oxygen bonding to a material and changing it. Rust is iron oxidizing. Atomic oxygen oxidizes spacecraft surfaces; here "corrosion" is the right word. *Rust, but in orbit, and fast.*

Kinetic energy / relative velocity — kinetic energy = energy of motion. Relative velocity = how fast two things approach each other; it doesn't matter which one is "really" moving. The s/c at 7.5 km/s hitting a still atom is the same as the atom hitting a still s/c at 7.5 km/s.

Ram-facing — the side of the spacecraft pointing in the direction of travel, which takes all the atomic-oxygen hits. *The windshield.*

Protected vs exposed — the slide photo's labels: a sample with a coating vs. one bare, after 10 months and 5.8 years in orbit. The bare one is shredded.

Layer / region — a slice of the atmosphere defined by how temperature behaves there. The boundaries are where the temperature trend flips. *Floors of a building.*

Stratosphere — 11–50 km. Calm, layered, holds the ozone layer, warms with height. *"Strato" = layered.*

Thermosphere — 90–500 km. Extremely hot by thermometer, almost no air. The ISS and most LEO satellites fly inside it. *"Thermo" = heat.*

Homosphere / heterosphere — homo = the bottom ~100 km where gases are stirred into one uniform mix. Hetero = above that, where gases separate by weight. *Smoothie vs. a layered cocktail.*

Density — how much stuff is packed into a space. **Numerical density (m^{-3})** on the chart = how many molecules per cubic meter. *People per square meter in a crowd.*

Constituent / component — one of the gases the air is made of. *Ingredients.*

O, N, H, He — single atoms of oxygen, nitrogen, hydrogen, and helium gas. The "atomic" forms. Up high, radiation breaks the paired molecules into these singles. *Loose Lego bricks instead of snapped-together pairs.*

Anomalous O — a tiny extra population of unusually energetic oxygen atoms that the chart's model adds above ~300 km. Minor; not something you need to remember beyond "it's still oxygen."

F10.7 / Ap — two numbers on the chart title describing how active the Sun and Earth's magnetic field were for the model run. The atmosphere puffs up when the Sun is active. Ignore the values.

Reactive — eager to bond chemically with other stuff. *Sticky.*

Erosion — material being worn away bit by bit. Atomic oxygen erodes polymers; the slide photo shows a surface roughened like sandblasted glass.

eV (electron-volt) — a tiny unit of energy used for single atoms and bonds. A chemical bond is a few eV; an atomic-oxygen hit at 7.5 km/s carries ~5 eV. *Enough to snap the bond.*

Kapton — a gold-colored polymer film used everywhere on spacecraft for insulation. Famous for being eaten by atomic oxygen unless coated.

S/C — shorthand for spacecraft, used on the slide.